Machine Learning - Undergraduate Level Assessment 25 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following are the spatial clustering algorithms?
   (a) Partitioning based clustering
   (b) K-means clustering
   (c) Grid based clustering
   (d) All of the above

2. Which of the following is NOT supervised learning?
   (a) PCA
   (b) Decision Tree
   (c) Linear Regression
   (d) Naive Bayesian

3. Statement 1| Highway networks were introduced after ResNets and eschew max pooling in favor of convolutions. Statement 2| DenseNets usually cost more memory than ResNets.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

4. Consider the Bayesian network given below. How many independent parameters would we need if we made no assumptions about independence or conditional independence H -> U <- P <- W?
   (a) 3
   (b) 4
   (c) 7
   (d) 15

5. Which of the following statements about Naive Bayes is incorrect?
   (a) Attributes are equally important.
   (b) Attributes are statistically dependent of one another given the class value.
   (c) Attributes are statistically independent of one another given the class value.
   (d) Attributes can be nominal or numeric

True / False (10 marks)

Answer True or False

6. True or False: Both Bayesians and frequentists agree on the use of prior distributions in probabilistic modeling for regression.
7. True or False: Reinforcement learning is often considered the most advanced approach in Yann LeCun's framework for machine learning.
8. True or False: Residual connections are commonly utilized in both ResNets and Transformers to facilitate training deep learning models.
9. True or False: The variance of the Maximum A Posteriori (MAP) estimate is the same as that of the Maximum Likelihood Estimate (MLE).
10. True or False: The ResNet-50 model has over 1 billion parameters, making it one of the largest neural network architectures.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the maximum total path sum in the given triangle.
12. Write a function that matches a string that has an a followed by three 'b'.

End of Paper